Physicochemical Differences and Latent Structure in Portuguese Vinho Verde Wines: A Multivariate Analysis Using MANOVA, PCA, and EFA

Ivan C. Cobarrubias

APM1220 – Applied Multivariate Data Analysis

Summative Assessment 1

1 Introduction

Wine quality is determined both by its chemical composition and by the sensory perception it elicits. Measurable physicochemical parameters, including acidity, sugar content, sulfur dioxide concentration, alcohol, and density, exert a direct influence on a wine's organoleptic profile, its stability, and its shelf life, and are consequently monitored by producers at successive stages of production. Because these parameters are recorded jointly and frequently exhibit mutual chemical dependence, they constitute a data structure well suited to multivariate techniques that model them simultaneously rather than in isolation.

This study examines the physicochemical characteristics of red and white Portuguese Vinho Verde wines by means of three analytical procedures. Multivariate analysis of variance is employed to determine whether the two wine types differ significantly in their overall chemical profile. Principal component analysis is applied to assess whether the 11 measured variables can be represented by a smaller number of dimensions. Exploratory factor analysis is used to investigate whether latent factors account for the correlations observed among these variables. The objective is not merely the execution of each procedure, but also the verification of its underlying assumptions and the substantive interpretation of the resulting solutions within the framework of wine chemistry.

2 Dataset

The present analysis employs the Wine Quality Dataset, obtained from the UCI Machine Learning Repository (Cortez et al., 2009b), which comprises physicochemical and sensory measurements recorded for red and white Vinho Verde wines produced in northern Portugal (Cortez et al., 2009a). The consolidated dataset contains 6,497 observations, of which

1,599 correspond to red wines and 4,898 to white wines. Each observation is described by 11 continuous physicochemical variables (fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol), together with a sensory quality score ranging from 0 to 10, assigned on the basis of evaluations by wine experts. An additional `color` variable, indicating whether a given wine is red or white, was introduced in order to identify the source file from which each observation was drawn. The data are distributed under a CC BY 4.0 license and are available at https://archive.ics.uci.edu/dataset/186/wine+quality.

3 Research Questions and Hypotheses

Research Question 1 (MANOVA)

Do red and white wines differ significantly in their combined physicochemical characteristics, as measured by fixed acidity, volatile acidity, residual sugar, chlorides, total sulfur dioxide, sulphates, and pH?

Let $\boldsymbol{\mu}_{\text{red}}$ and $\boldsymbol{\mu}_{\text{white}}$ denote the population mean vectors of these seven variables for red and white wines, respectively.

$$H_0 : \boldsymbol{\mu}_{\text{red}} = \boldsymbol{\mu}_{\text{white}} \quad \text{(the mean vector is the same for red and white wines)}$$
$$H_1 : \boldsymbol{\mu}_{\text{red}} \neq \boldsymbol{\mu}_{\text{white}} \quad \text{(the mean vectors differ in at least one variable)}$$

Research Question 2 (PCA)

Can the 11 physicochemical measurements be summarized by a smaller number of principal components that retain a substantial proportion of the total variability?

PCA is a descriptive technique and has no formal hypothesis test. Because several variables are expected to be strongly correlated, a small number of components is expected to capture most of the variance.

Research Question 3 (EFA)

What underlying latent dimensions or factors explain the relationships among the physicochemical characteristics of wine?

The testable hypothesis concerns factorability (Bartlett's test of sphericity):

$H_0 : \mathbf{R} = \mathbf{I}$ (the correlation matrix is an identity matrix; the variables are uncorrelated)
$H_1 : \mathbf{R} \neq \mathbf{I}$

The Kaiser–Meyer–Olkin (KMO) measure is used alongside it to judge sampling adequacy.

4 Methodology

4.1 Data Preparation

The red and white wine files are semicolon-delimited and were imported separately by means of the `read_delim()` function from the *readr* package, with a period employed as the decimal mark. The two datasets were subsequently consolidated into a single analytical dataset, and a categorical variable `color` ("red" or "white") was derived from the source file of each observation. Column names were standardized by replacing spaces with underscores. The structure of the combined data was then inspected with respect to variable types and the number of observations and variables, and the data were examined for missing values and for duplicate observations, the latter defined as rows that are identical across all variables, including `color`.

All duplicate rows were retained because the files contain no wine identifier. Since there were no identifiers, it was not safe to assume that the duplicate observations were the same entries therefore duplicates were treated as separate observations with the same input values.

4.2 Exploratory Data Analysis

Descriptive statistics (mean, standard deviation, median, minimum, maximum, skewness, and kurtosis) were computed for the 11 physicochemical variables, overall and separately by wine color, using the *psych* package. Because the variables are measured on very different scales, relative variability was compared using the coefficient of variation ($CV = SD/M$). Distributions were examined with density plots by wine color, and red and white wines were compared using boxplots. Associations among the variables were summarized with a Pearson correlation matrix and heatmap, and with a scatterplot matrix of six selected variables drawn from a random sample of 1,500 wines (seed = 123) to limit overplotting.

Univariate outliers were identified for each variable using the 1.5 × interquartile range (IQR) rule. Multivariate outliers were identified using the Mahalanobis distance computed on all 11 physicochemical variables, with observations flagged when the squared distance exceeded the chi-square critical value for 11 degrees of freedom at $p < .001$ ($\chi^2_{.999}(11) = 31.26$). Flagged observations were examined but retained, because extreme values in wine chemistry are plausible measurements rather than evident errors." All analyses were conducted in R version 4.4.0 using the *tidyverse* and *psych* packages.

4.3 MANOVA

Dependent variable selection

A one-way multivariate analysis of variance (MANOVA) tested whether red and white wines differ in their combined physicochemical characteristics. Wine color (red vs. white) was the independent variable. The analysis included all 6,497 wines, retaining duplicate

rows and flagged outliers (Sections 4.1 and 4.2). Dependent variables were selected from 11 physicochemical measures using three criteria.

First, each variable required a large standardized between-group difference (Cohen's $|d| \geq 0.8$). Second, variables were required to reflect distinct chemical aspects rather than redundant measures. Third, redundancy was assessed using Pearson correlations and VIFs: for pairs with $|r| > .70$, only the variable with the larger group difference was retained; variables with VIFs > 10 were excluded. VIFs were then recomputed and the final set required VIFs below 5.

This process produced seven dependent variables: fixed acidity, volatile acidity, and pH (acidity); residual sugar (sugar); chlorides and sulphates (salts); and total sulfur dioxide (preservatives). Free sulfur dioxide was excluded as redundant with total sulfur dioxide. Density was excluded because it is largely determined by alcohol and residual sugar. Alcohol and citric acid were not included due to small or nonsignificant group differences. The sensory quality score was not analyzed per the assessment instructions. Evidence for each decision is reported in Section 5.2.

Assumption checks

MANOVA assumptions were assessed prior to interpretation. Independence was evaluated using the study design. Multivariate normality was examined separately within each wine type using Mardia's tests and chi-square Q–Q plots of squared Mahalanobis distances; graphical and practical departures were considered given the sensitivity of formal tests in large samples. Multivariate outliers were identified within each wine type using Mahalanobis distance, with observations flagged when the squared distance exceeded the chi-square critical value for seven degrees of freedom at $p < .001$.

Homogeneity of covariance matrices was tested with Box's M test at $\alpha = .001$, motivated by the test's sensitivity with large samples and unequal group sizes (1,599 red and 4,898 white). Interrelationships among dependent variables were evaluated using within-group correlation matrices, the residual correlation matrix from the MANOVA model, VIFs, the determinant of the residual correlation matrix, and Bartlett's test of sphericity applied to the residual correlation matrix.

Multivariate test

MANOVA was conducted in R using the `manova()` function at $\alpha = .05$. Pillai's trace was the primary statistic due to its relative robustness to violations of multivariate normality and covariance homogeneity under unequal group sizes. Wilks' lambda, Hotelling's trace, and Roy's largest root were computed for comparison. For each statistic, the test value, approximate F value, degrees of freedom, and p-value were recorded. Partial η^2 was used as the multivariate effect size (equal to Pillai's trace, and $1 - \Lambda$, in the two-group case) and interpreted using .01, .06, and .14 as benchmarks.

Follow-up analyses

When the omnibus MANOVA was significant, follow-up univariate ANOVAs were conducted for each of the seven dependent variables. A Bonferroni-adjusted level of $\alpha = .05/7 \approx .0071$ controlled the familywise error rate. Effect sizes were reported as η^2 and Cohen's d (0.2, 0.5, 0.8 for small, medium, and large effects). Welch t-tests were included as robustness checks for unequal variances. With only two groups, no additional post-hoc procedures were required. Discriminant analysis was used to provide standardized discriminant coefficients and structure correlations that reflect each variable's contribution to separating red and white wines after accounting for the others.

4.4 Principal Component Analysis

Correlation structure and standardization

Principal component analysis (PCA) was applied to the 11 physicochemical variables (fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol) for all 6,497 wines, with duplicate rows and flagged outliers retained as described earlier. Wine color and the sensory quality score were not included. Before extracting components, the correlation structure was examined to determine whether the variables are related enough for PCA to be worthwhile. The correlation matrix was inspected, the number of variable pairs with at least moderate correlations ($|r| \geq .30$ and $|r| \geq .50$) was counted, and two standard checks were computed: the Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy, both overall and for each variable (values above .60 were considered acceptable and above .80 good, and individual values below .50 as problematic), and Bartlett's test of sphericity, which tests the null hypothesis that the correlation matrix is an identity matrix.

Because the variables are measured in different units and on very different scales (for example, total sulfur dioxide has a variance thousands of times larger than density), each variable was standardized to a mean of 0 and a standard deviation of 1 before the analysis. This is equivalent to performing the PCA on the correlation matrix. Standardization ensures that every variable contributes on an equal footing, whereas an unstandardized analysis would be dominated by the variables with the largest variances, regardless of how much structure they actually share with the others.

Component retention

The components were extracted with the `prcomp()` function in R. The number of components to retain was determined by weighing several sources of evidence together instead of relying on any single rule: the eigenvalues, the scree plot (looking for the point at which the curve levels off), the proportion of variance explained by each component, the cumulative proportion of variance explained (using 60–70% as a general guide), and parallel analysis. Parallel analysis compares the observed eigenvalues with those obtained from random data of the same size, and a component is retained only if its observed

eigenvalue exceeds the corresponding value from the random data; it was run with the `fa.parallel()` function of the *psych* package using 100 iterations and a fixed random seed (123) for reproducibility. The eigenvalue-greater-than-one (Kaiser) criterion was reported for reference but was not used as the deciding rule. The final number of components also took into account whether the retained components could be given a meaningful chemical interpretation.

Component loadings and interpretation

For the retained components, the component loadings, defined as the correlations between each variable and each component, were computed from the eigenvectors scaled by the square roots of the eigenvalues. Loadings with an absolute value of .40 or higher were treated as substantial. To determine which variables contribute most strongly to each component, the percentage contribution of each variable (the squared eigenvector coefficient, which sums to 100% within a component) was also calculated, along with the communality of each variable, that is, the proportion of its variance reproduced by the retained components. Because the sign of a component is arbitrary, components were interpreted in terms of contrasts between groups of variables (variables with loadings of the same sign vary together and those with opposite signs vary in opposite directions) rather than in terms of overall direction.

Each retained component was named according to the variables with substantial loadings and the chemical characteristic they have in common. The results were visualized with a scree plot, loading plots of the variables for pairs of the first components, and a biplot of the wine scores for the same pairs of components with the points colored by wine color, so that any separation between red and white wines along the components is visible.

4.5 Exploratory Factor Analysis

Factorability

Exploratory factor analysis (EFA) was applied to the same 11 physicochemical variables and 6,497 wines used in the PCA; wine color and the sensory quality score were excluded. Factorability was assessed with the correlation matrix, the Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy (overall and per variable; values above .60 were considered acceptable, above .80 good, and below .50 problematic), Bartlett's test of sphericity, and the determinant of the correlation matrix. Because the KMO was low in the PCA stage, it was also recomputed within each wine color and with density removed, to see whether the pooling of red and white wines or the redundancy of density with alcohol and residual sugar accounts for the low values. These recomputations were diagnostic only, and the factor model was fitted to all 11 variables.

Number of factors

The number of factors was determined by weighing several sources of evidence together: the scree plot, parallel analysis for common factors (`fa.parallel()` with `fa = "fa"`, 100 simulated data sets, the 95th percentile of the random eigenvalues as the threshold, and seed 123), the eigenvalues of the correlation matrix, Velicer's minimum average partial (MAP) test, the fit of solutions with different numbers of factors (RMSEA and BIC), and the interpretability of each solution. Solutions with one fewer and one more factor than the chosen number were also examined to check that the chosen solution gives the clearest simple structure.

Extraction

Factors were extracted by the minimum residual (minres) method with the `fa()` function of the `psych` package. EFA models only the variance that variables share (the common variance) and treats the remainder as unique variance, unlike PCA, which analyzes total variance. Minimum residual extraction does not assume multivariate normality, which is appropriate here because multivariate normality was rejected in the MANOVA diagnostics (Section 5.3) and several variables are strongly skewed (Section 5.1), and it is less prone to improper solutions than maximum likelihood. Maximum likelihood extraction was used as a check on the stability of the solution.

Rotation

Factors were rotated with the oblique oblimin rotation (`GPArotation`). Chemical characteristics of wine (acidity, sugar and alcohol content, preservation) are not independent in practice, so the underlying factors were expected to be correlated, and an oblique rotation allows this without forcing it: if the factors turn out to be uncorrelated, an oblique solution reduces to an orthogonal one. The factor correlation matrix was examined to confirm the choice, and promax rotation was used as a check on the solution.

Interpretation of the solution

The pattern matrix was used to interpret the factors. Loadings of |.40| or higher were treated as salient. A variable was considered cross-loading when it had loadings of |.30| or higher on more than one factor, and weak when it had no loading of |.40| or higher. Communalities (the proportion of each variable's variance explained by the common factors) and uniquenesses (their complements) were examined, and communalities of 1 or greater were checked as signs of an improper (Heywood) solution. Model fit was described with the root mean square residual (RMSR), RMSEA, and Tucker–Lewis index, as descriptive indices because they are sensitive to sample size. Each factor was named according to the variables that load on it and the chemical characteristic they have in common. Factor scores were computed by the regression method, and the standardized mean difference

between red and white wines on each factor was calculated to aid interpretation; wine color was not used in the extraction.

Robustness

The stability of the solution was assessed by comparing it with solutions obtained with promax rotation, with maximum likelihood extraction, and after excluding the multivariate outliers flagged in Section 4.2, using Tucker's congruence coefficient between corresponding factors (values of .95 or higher were taken to indicate essentially identical factors; Lorenzo-Seva & ten Berge, 2006). Because the data pool two wine types with different chemical profiles, the factor solution was also fitted separately to the red and white wines to see whether the pooled structure holds within each type.

4.6 Software

All analyses were conducted in R version 4.4.0 in a Kaggle notebook. Data manipulation and visualization used the *tidyverse* suite (version 2.0.0), including *readr*, *dplyr*, *tidyr*, and *ggplot2*. Descriptive statistics, the Kaiser–Meyer–Olkin measure, Bartlett's test, parallel analysis, and exploratory factor analysis were computed with the *psych* package, Box's M test with *heplots*, and oblique factor rotation with *GPArotation*. MANOVA and principal component analysis used base R functions (`manova()` and `prcomp()`). The report was typeset in LaTeX. The analysis code, data, figures, and tables are available at https://github.com/Nedevi/APM1220-SA1-Applied-Multivariate-Data-Analysis.

5 Results

5.1 Exploratory Data Analysis

Data structure and quality

The combined dataset contained 6,497 observations on 13 variables: the 11 physicochemical variables, the sensory quality score, and wine color. Of these wines, 1,599 (24.6%) were red and 4,898 (75.4%) were white, so the two groups are clearly unbalanced. There were no missing values in any variable. A total of 1,177 rows (18.1%) were duplicates, comprising 240 red wines (15.0% of red wines) and 937 white wines (19.1% of white wines).

Descriptive statistics and variability

Table 1 presents descriptive statistics for the 11 physicochemical variables. Because the variables are on different scales, relative variability is best compared through the coefficient of variation. Residual sugar was the most variable variable ($CV = 0.87$), followed by chlorides ($CV = 0.63$), free sulfur dioxide ($CV = 0.58$), total sulfur dioxide ($CV = 0.49$), volatile acidity ($CV = 0.49$), and citric acid ($CV = 0.46$). In contrast,

density ($CV = 0.003$), pH ($CV = 0.05$), alcohol ($CV = 0.11$), and fixed acidity ($CV = 0.18$) varied relatively little across wines.

Table 1: Descriptive Statistics for the 11 Physicochemical Variables (N = 6,497)

Variable	*M*	*SD*	*Mdn*	Min	Max	Skew	Kurtosis	CV
Fixed acidity	7.215	1.296	7.000	3.800	15.900	1.722	5.054	0.180
Volatile acidity	0.340	0.165	0.290	0.080	1.580	1.494	2.820	0.485
Citric acid	0.319	0.145	0.310	0.000	1.660	0.472	2.393	0.456
Residual sugar	5.443	4.758	3.000	0.600	65.800	1.435	4.353	0.874
Chlorides	0.056	0.035	0.047	0.009	0.611	5.397	50.841	0.625
Free sulfur dioxide	30.525	17.749	29.000	1.000	289.000	1.220	7.896	0.581
Total sulfur dioxide	115.745	56.522	118.000	6.000	440.000	−0.001	−0.373	0.488
Density	0.995	0.003	0.995	0.987	1.039	0.503	6.597	0.003
pH	3.219	0.161	3.210	2.720	4.010	0.387	0.365	0.050
Sulphates	0.531	0.149	0.510	0.220	2.000	1.796	8.643	0.280
Alcohol	10.492	1.193	10.300	8.000	14.900	0.565	−0.533	0.114

Note. CV = coefficient of variation (SD/M). Skew and kurtosis are the sample skewness and excess kurtosis.

Distributions

Figure 1 shows the distributions of the variables by wine color. Several variables were strongly right-skewed, most notably chlorides (skewness = 5.40, kurtosis = 50.84), sulphates (1.80), fixed acidity (1.72), volatile acidity (1.49), and residual sugar (1.44). Free sulfur dioxide was also right-skewed (1.22) with heavy tails (kurtosis = 7.90). Total sulfur dioxide was approximately symmetric overall (skewness ≈ 0), although this reflects a mixture of two groups with very different centers, and alcohol and pH were only mildly skewed. Density had a high kurtosis (6.60), driven by an extreme maximum of 1.039 against a median of 0.995.

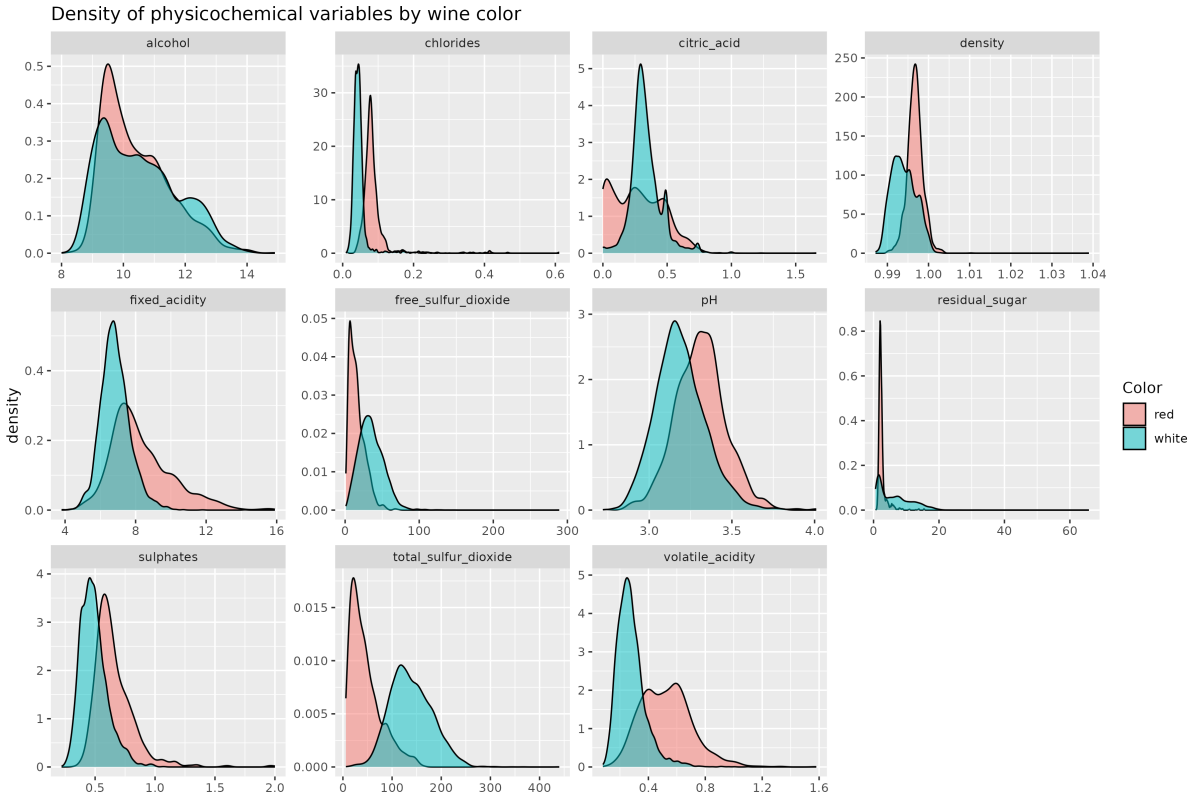


Figure 1: Density plots of the 11 physicochemical variables by wine color.

Red versus white wines

Red and white wines differed noticeably on most variables (Table 2; Figure 2). Red wines had higher fixed acidity ($M = 8.32$ vs. 6.85), volatile acidity (0.53 vs. 0.28), chlorides (0.087 vs. 0.046), sulphates (0.66 vs. 0.49), pH (3.31 vs. 3.19), and density (0.9967 vs. 0.9940). White wines had higher residual sugar (6.39 vs. 2.54), free sulfur dioxide (35.31 vs. 15.87), total sulfur dioxide (138.36 vs. 46.47), and citric acid (0.33 vs. 0.27). The separation was especially marked for total sulfur dioxide, where the group means differ by more than two standard deviations. Alcohol was nearly identical between colors (10.42 vs. 10.51). White wines also showed greater spread in residual sugar ($SD = 5.07$ vs. 1.41) and in free and total sulfur dioxide.

Table 2: Means and Standard Deviations of the Physicochemical Variables by Wine Color

Variable	Red (n = 1,599) M (SD)	White (n = 4,898) M (SD)
Fixed acidity	8.32 (1.74)	6.85 (0.84)
Volatile acidity	0.53 (0.18)	0.28 (0.10)
Citric acid	0.27 (0.19)	0.33 (0.12)
Residual sugar	2.54 (1.41)	6.39 (5.07)
Chlorides	0.087 (0.047)	0.046 (0.022)
Free sulfur dioxide	15.87 (10.46)	35.31 (17.01)
Total sulfur dioxide	46.47 (32.90)	138.36 (42.50)
Density	0.9967 (0.0019)	0.9940 (0.0030)
pH	3.31 (0.15)	3.19 (0.15)
Sulphates	0.66 (0.17)	0.49 (0.11)
Alcohol	10.42 (1.07)	10.51 (1.23)

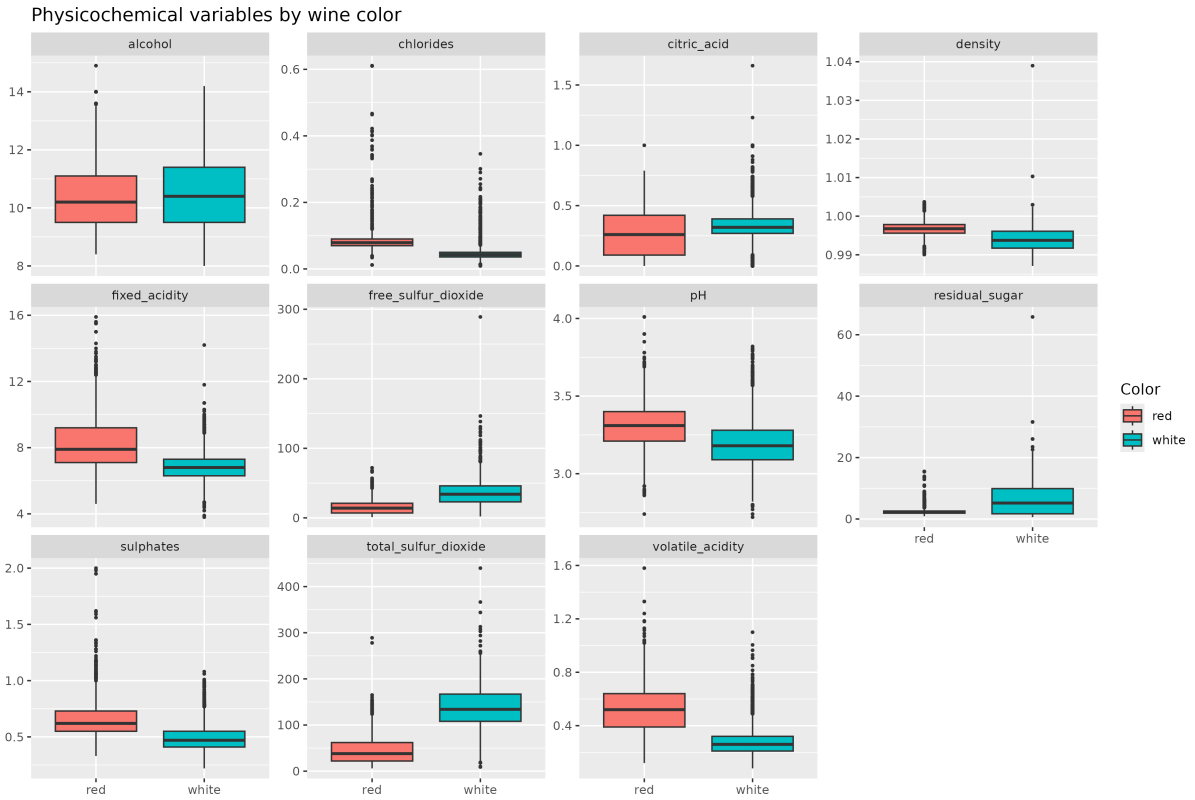


Figure 2: Boxplots of the 11 physicochemical variables by wine color.

Correlations

Figure 3 displays the correlation matrix. The strongest association was between free and total sulfur dioxide (r = .72), followed by a negative relationship between density and alcohol (r = −.69). Density was also positively related to residual sugar (r = .55) and fixed acidity (r = .46), and total sulfur dioxide was positively related to residual

sugar ($r = .50$). Moderate associations were also present between volatile acidity and total sulfur dioxide ($r = -.41$), chlorides and sulphates ($r = .40$), residual sugar and free sulfur dioxide ($r = .40$), and volatile acidity and citric acid ($r = -.38$). These correlations were computed on the pooled data, so some of them may partly reflect the red–white differences described above rather than relationships within each wine type.

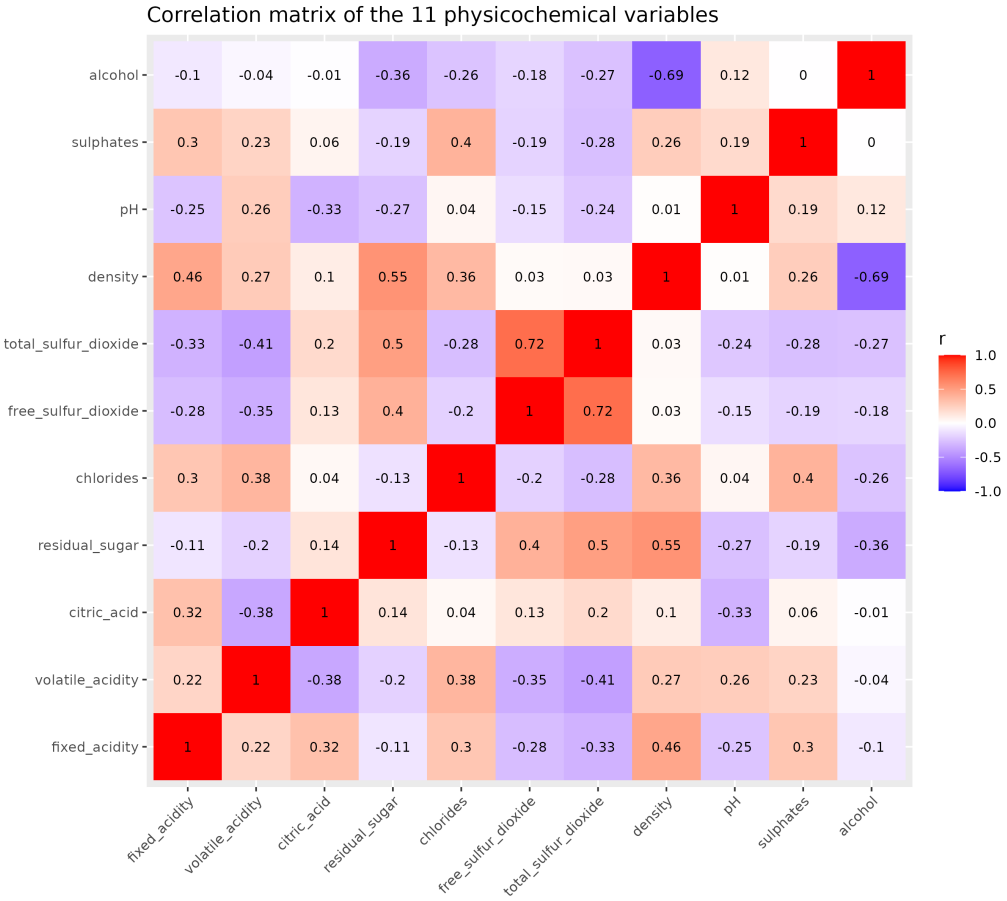


Figure 3: Correlation matrix of the 11 physicochemical variables (Pearson r, combined red and white wines).

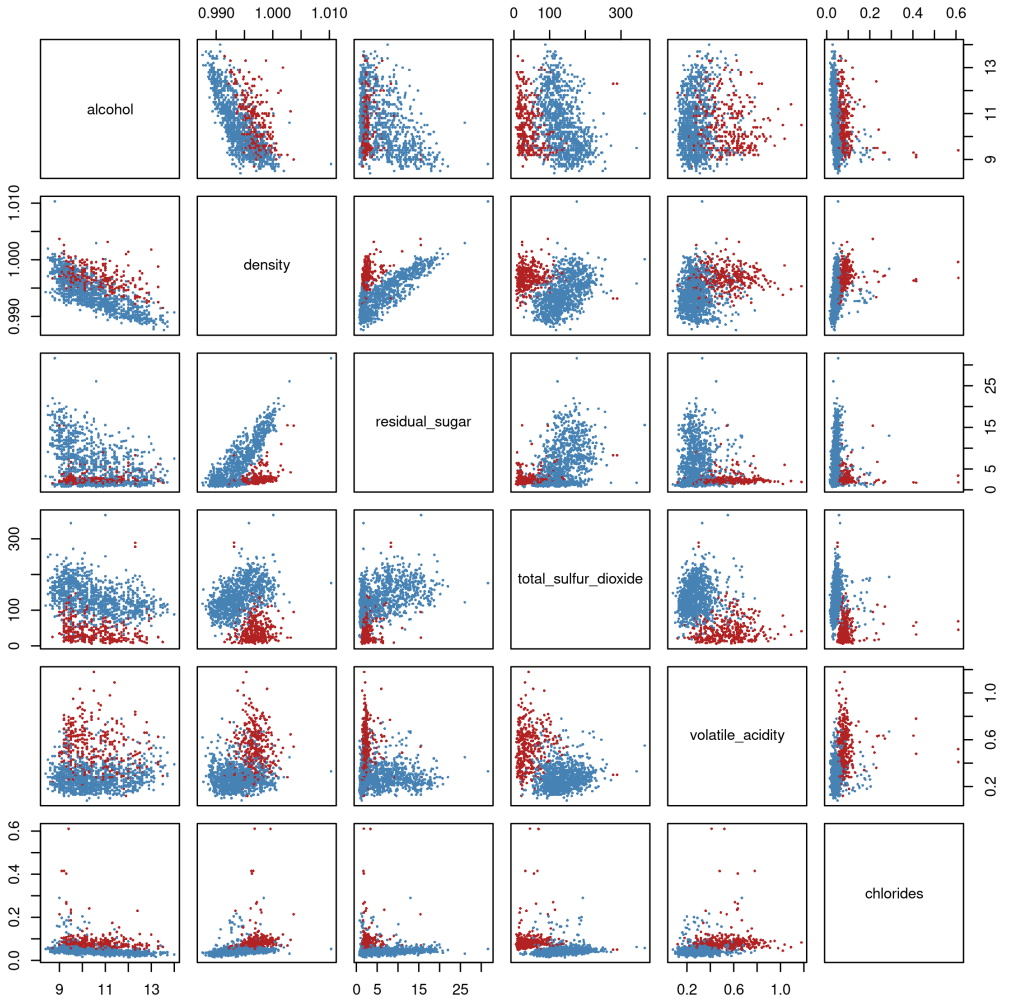


Figure 4: Scatterplot matrix of six physicochemical variables for a random sample of 1,500 wines (red = dark red, white = blue).

Outliers and unusual observations

Using the 1.5 × IQR rule, the variables with the most flagged values were citric acid (509 wines, 7.8%), volatile acidity (377, 5.8%), fixed acidity (357, 5.5%), chlorides (286, 4.4%), and sulphates (191, 2.9%). Very few values were flagged for density (3), alcohol (3), and total sulfur dioxide (10). Extreme individual values included a residual sugar of 65.8, a density of 1.039, and a total sulfur dioxide of 440. On the multivariate criterion, 175 wines (2.7%) exceeded the Mahalanobis distance cutoff and were therefore flagged as multivariate outliers. These observations were retained for the main analyses.

Considerations before PCA and EFA

Several features of the data bear on the later analyses. First, the variables are on very different scales (e.g., total sulfur dioxide has a mean of 115.7 while chlorides has a mean of 0.056), so the variables must be standardized before PCA. Second, the strong skewness

and heavy tails in chlorides, residual sugar, sulphates, and free sulfur dioxide, together with the flagged outliers, may influence the components and factors, since correlation-based methods are sensitive to extreme values. Third, free and total sulfur dioxide ($r = .72$) and density with alcohol ($r = -.69$) are strongly correlated and are likely to load on the same dimensions. Finally, because the data pool two wine types with different chemical profiles, the correlation structure used for PCA and EFA is a mixture and may partly reflect the red–white distinction.

5.2 MANOVA

Dependent variable selection

Table 3 summarizes the evidence used to select the dependent variables from the 11 physicochemical variables. Seven variables met the criteria and were retained: total sulfur dioxide ($d = -2.28$), volatile acidity ($d = 2.00$), chlorides ($d = 1.39$), sulphates ($d = 1.30$), fixed acidity ($d = 1.29$), residual sugar ($d = -0.86$), and pH ($d = 0.81$). Free sulfur dioxide was excluded because of its correlation with total sulfur dioxide ($r = .72$), density because of its very high variance inflation factor (VIF = 15.9), and alcohol ($d = -0.08$) and citric acid ($d = -0.44$) because of small group differences. The VIFs of residual sugar (7.2) and fixed acidity (4.9) in Table 3 were inflated by density; with density removed, no selected variable had a VIF above 5 in the pooled data, and within wine type the VIFs ranged from 1.06 to 1.42.

Table 3: Evidence for the Selection of Dependent Variables for the MANOVA

Variable	Cohen's d	Largest $\lvert r \rvert$ (with)	VIF	Decision
Total sulfur dioxide	−2.28	.72 (free SO_2)	2.9	Retained
Volatile acidity	2.00	.41 (total SO_2)	1.9	Retained
Chlorides	1.39	.40 (sulphates)	1.6	Retained
Sulphates	1.30	.40 (chlorides)	1.5	Retained
Fixed acidity	1.29	.46 (density)	4.9	Retained
Free sulfur dioxide	−1.24	.72 (total SO_2)	2.1	Excluded (redundant with total SO_2)
Density	0.99	.69 (alcohol)	15.9	Excluded (VIF > 10)
Residual sugar	−0.86	.55 (density)	7.2	Retained
pH	0.81	.33 (citric acid)	2.5	Retained
Citric acid	−0.44	.38 (volatile acidity)	1.6	Excluded ($\lvert d \rvert < 0.8$)
Alcohol	−0.08	.69 (density)	4.8	Excluded ($\lvert d \rvert < 0.8$)

Note. Cohen's d is the standardized mean difference (red − white). Largest $\lvert r \rvert$ and VIF were computed from the pooled data for all 11 variables.

Assumption checks

Independence

The data are observational measurements of individual wines, and the files contain no producer, batch, or vintage identifiers. Wines from the same producer or batch could therefore be more alike than wines chosen at random, and duplicate rows could not be verified as distinct samples. Independence of observations could thus not be confirmed and is treated as a limitation.

Multivariate outliers

Using the Mahalanobis distance within each wine type, 53 red wines (3.3%) and 117 white wines (2.4%) exceeded the critical value of $\chi^2_{.999}(7) = 24.32$. These observations were retained, as they are plausible measurements rather than evident errors.

Multivariate normality

Mardia's tests indicated significant departures from multivariate normality in both wine types. For red wines, multivariate skewness was 70.37, $\chi^2(84) = 18{,}753.69$, $p < .001$, and multivariate kurtosis was 164.13 ($z = 180.12$, $p < .001$), compared with an expected value of 63 under normality. For white wines, skewness was 35.34, $\chi^2(84) = 28{,}845.33$, $p < .001$, and kurtosis was 123.16 ($z = 187.54$, $p < .001$). The chi-square Q–Q plots (Figure 5) agree: the points follow the reference line only in the lower range and then curve sharply upward, with the largest squared distances (about 145 for red and 230 for white wines) far beyond what would be expected under normality (about 30). The data therefore have heavy right tails, so the assumption is not met. Because the test of the group difference rests on large samples, only two groups, and Pillai's trace, which is the most robust of the multivariate statistics, the departure was treated as a limitation rather than a barrier to the analysis.

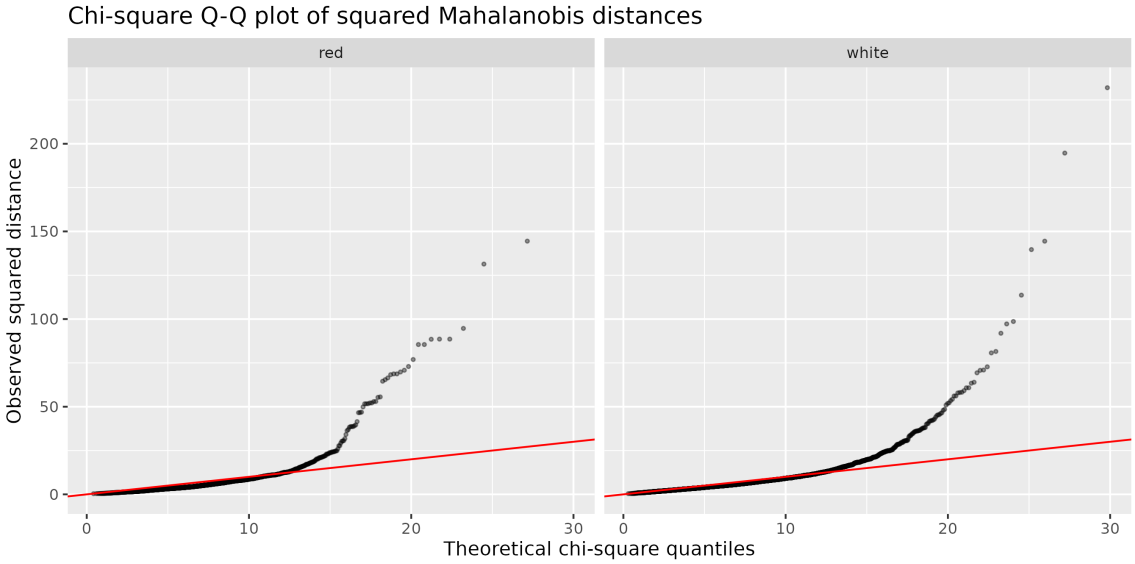


Figure 5: Chi-square Q–Q plots of the squared Mahalanobis distances of the seven dependent variables, by wine color. The red line marks the expected relationship under multivariate normality.

Homogeneity of covariance matrices.

Box's M test indicated that the covariance matrices differed between red and white wines, $\chi^2(28) = 7{,}533.9$, $p < .001$. Because the test is very sensitive in large samples, the matrices were also compared directly, and they differed considerably, not merely by a small amount. White wines showed much larger variances for residual sugar (25.73 vs. 1.99) and total sulfur dioxide (1,806.09 vs. 1,082.10), whereas red wines showed larger variances for fixed acidity (3.03 vs. 0.71), volatile acidity (0.032 vs. 0.010), chlorides (0.0022 vs. 0.0005), and sulphates (0.029 vs. 0.013); the variance of pH was similar (0.024 vs. 0.023). The assumption is therefore not met. Since the larger group has the larger variance for some variables and the smaller group for others, the net effect on the test is hard to predict, which is why Pillai's trace was used and why Welch tests were added as a robustness check in the follow-up analyses. This is also treated as a limitation.

Relationships among the dependent variables.

The dependent variables were related but not redundant. Within each wine type, all correlations were below $|r| = .70$; the largest was between fixed acidity and pH in red wines ($r = -.68$), and few others exceeded .40 (fixed acidity and pH in white wines, $r = -.43$; residual sugar and total sulfur dioxide in white wines, $r = .40$). The correlations of the MANOVA residuals were similarly modest, with a maximum of $|r| = .50$ (fixed acidity and pH). Bartlett's test of sphericity on the residual correlation matrix was significant, $\chi^2(21) = 4{,}168.80$, $p < .001$, indicating that the variables are sufficiently related to justify a multivariate analysis. Within-group VIFs ranged from 1.06 to 1.42, and the determinant of the residual correlation matrix was 0.53, far above the threshold of .00001 for multicollinearity.

Multivariate test

The MANOVA showed a significant multivariate effect of wine color on the seven physicochemical variables considered together, Pillai's trace = 0.79, $F(7, 6489) = 3518.70$, $p < .001$, partial $\eta^2 = .79$ (Table 4). The effect is very large relative to the conventional benchmark of .14: wine color accounts for about 79% of the variance in the best linear combination of the dependent variables. Because there were only two groups, the four test statistics were equivalent and gave identical F values, degrees of freedom, and p-values, so the conclusion does not depend on the choice of statistic. We therefore reject the null hypothesis that the mean vectors are equal.

Table 4: Multivariate Tests of the Effect of Wine Color on the Seven Dependent Variables

Test	Statistic	F	df_1	df_2	p
Pillai's trace	0.791	3518.70	7	6489	$< .001$
Wilks' Λ	0.209	3518.70	7	6489	$< .001$
Hotelling's trace	3.796	3518.70	7	6489	$< .001$
Roy's largest root	3.796	3518.70	7	6489	$< .001$

Note. Partial $\eta^2 = 1 - \Lambda = .79$.

Follow-up analyses

Follow-up univariate ANOVAs showed that red and white wines differed significantly on every one of the seven variables, all with $p < .001$ both before and after the Bonferroni adjustment ($\alpha = .0071$), and the Welch tests gave the same conclusion (Table 5). The effect sizes, however, varied widely (Figure 6). Wine color explained the most variance in total sulfur dioxide ($\eta^2 = .49$, $d = -2.28$) and volatile acidity ($\eta^2 = .43$, $d = 2.00$), followed by chlorides ($\eta^2 = .26$), sulphates ($\eta^2 = .24$), and fixed acidity ($\eta^2 = .24$), with residual sugar ($\eta^2 = .12$) and pH ($\eta^2 = .11$) showing the smallest, though still medium to large, effects. White wines had higher total sulfur dioxide and residual sugar, whereas red wines had higher volatile acidity, chlorides, sulphates, fixed acidity, and pH.

Table 5: Follow-up Univariate ANOVAs for the Seven Dependent Variables

Variable	$F(1, 6495)$	η^2	Cohen's d	Higher in
Total sulfur dioxide	6253.0	.49	−2.28	White
Volatile acidity	4829.0	.43	2.00	Red
Chlorides	2316.0	.26	1.39	Red
Sulphates	2022.0	.24	1.30	Red
Fixed acidity	2017.0	.24	1.29	Red
Residual sugar	899.8	.12	−0.86	White
pH	789.0	.11	0.81	Red

Note. All $p < .001$ (unadjusted, Bonferroni-adjusted, and Welch). Cohen's d is the standardized difference (red − white), so a positive value means the mean is higher in red wines.

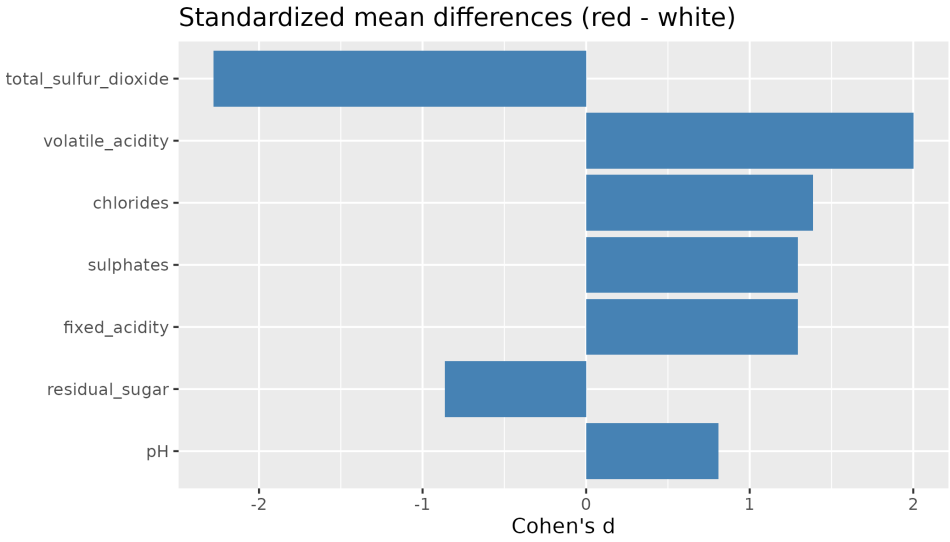


Figure 6: Standardized mean differences (Cohen's d, red − white) for the seven dependent variables.

Because univariate tests ignore the correlations among the variables, the standardized discriminant function coefficients and structure correlations were also examined to see how each variable contributes to separating the two wine types after accounting for the others (Table 6). Total sulfur dioxide (0.65), volatile acidity (−0.57), and fixed acidity (−0.57) made the largest unique contributions, followed by pH (−0.47), chlorides (−0.33), and sulphates (−0.31). Residual sugar contributed almost nothing once the other variables were taken into account (coefficient = −0.05), even though it differed clearly between wine types on its own (d = −0.86) and correlated .39 with the discriminant function; its separation of red from white wines is thus largely shared with the other variables, notably total sulfur dioxide, with which it is moderately correlated within wine type. Conversely, pH contributed substantially after accounting for fixed acidity, although it had the smallest univariate effect, which suggests that red wines have a higher pH than their fixed acidity alone would predict.

Table 6: Discriminant Function Coefficients for the Seven Dependent Variables

Variable	Standardized coefficient	Structure correlation
Total sulfur dioxide	0.648	0.787
Volatile acidity	−0.568	−0.734
Fixed acidity	−0.566	−0.547
pH	−0.465	−0.370
Chlorides	−0.325	−0.576
Sulphates	−0.307	−0.548
Residual sugar	−0.053	0.392

Note. Positive values indicate higher scores for white wines on the discriminant function; negative values indicate higher scores for red wines.

Interpretation

Red and white Vinho Verde wines differ significantly in their combined physicochemical characteristics, and the difference is large: the two wine types are largely separable on these seven measurements, with wine color accounting for about 79% of the variance in their best combination. The chemistry that separates them is two-sided. White wines are marked by high total sulfur dioxide, the preservative measure, and by higher residual sugar, whereas red wines are characterized by higher volatile acidity, fixed acidity, chlorides, sulphates, and pH.

The two variables that matter most are total sulfur dioxide and volatile acidity, which together carry the largest share of the difference both on their own (the two largest effect sizes) and after accounting for the other variables. Fixed acidity and pH reflect an acidity dimension in which red wines are both more acidic in their fixed acid content and higher in pH than expected from that content alone. Chlorides and sulphates, the mineral-salt measures, are higher in red wines and add more to the univariate differences than to the unique multivariate separation. Residual sugar differs between wine types but largely overlaps with the other variables in distinguishing them.

These conclusions should be interpreted with the unmet assumptions identified in the diagnostic checks in mind. Multivariate normality and equality of covariance matrices were not satisfied, independence could not be verified, and the data are observational. Despite these shortcomings, the magnitude of the observed effect, the concordance of all four multivariate test statistics, and the consistency between the Welch-adjusted tests and the conventional ANOVAs render it improbable that these violations materially alter the central inference that red and white wines are chemically distinct. Exact F statistics and associated p-values should nevertheless be interpreted with appropriate caution.

5.3 Principal Component Analysis

5.3.1 Correlation structure.

Of the 55 variable pairs, 16 had $|r| \geq .30$ and 3 had $|r| \geq .50$; the largest correlation was between free and total sulfur dioxide ($r = .72$). Bartlett's test rejected the hypothesis that the correlation matrix is an identity matrix, $\chi^2(55) = 37{,}094.62$, $p < .001$, so the variables are not mutually uncorrelated. The Kaiser–Meyer–Olkin measure, however, was low: overall MSA = .405, below the .50 floor and the .60 level set in the Methodology. Five variables fell below .50 individually (pH = .21, alcohol = .27, fixed acidity = .28, residual sugar = .29, density = .30), whereas free sulfur dioxide (.75), chlorides (.73), total sulfur dioxide (.72), citric acid (.62), volatile acidity (.61), and sulphates (.56) were acceptable.

The evidence on suitability is therefore mixed. Bartlett's test is nearly guaranteed to be significant at $N = 6{,}497$ and is weak evidence on its own. The low KMO indicates that, for many variables, the partial correlations remain large relative to the zero-order correlations; this is plausible for a set in which density is closely tied to alcohol and

residual sugar (Figure 3) and in which two chemically different wine types are pooled, although the present analysis does not isolate the cause. KMO is a stricter requirement for a common-factor model than for PCA, which is a descriptive re-expression of the total variance and does not assume latent factors. PCA was therefore retained, but the components are read as summaries of the observed variables rather than as evidence of latent constructs, and the low KMO is carried forward as a caution for the factor analysis (Section 5.5).

5.3.2 Component retention.

Because the variables were standardized, the PCA is equivalent to an eigen-decomposition of the correlation matrix, and the eigenvalues sum to 11. Table 7 and Figure 7 give the eigenvalues, variance explained, and the parallel-analysis thresholds (95th percentile of 100 random data sets). The first three components had eigenvalues of 3.03, 2.49, and 1.56, and each exceeded its parallel-analysis threshold (about 1.03 to 1.06). The fourth component (eigenvalue = 0.97) fell below both the parallel-analysis threshold (1.02) and the Kaiser value of 1, and the scree plot bends at this point, after which the eigenvalues decline gradually from 0.72 to 0.03. Parallel analysis, the scree plot, and the Kaiser rule thus agree on three components, so the decision does not rest on the eigenvalue-greater-than-one rule alone. The three components jointly explain 64.4% of the total variance, within the 60–70% guide. A fourth component would raise this to 73.2% but would not be distinguishable from components extracted from random data, and was not retained.

Table 7: Eigenvalues, Variance Explained, and Parallel-Analysis Thresholds for the Principal Components

Component	Eigenvalue	% Variance	Cumulative %	PA threshold	> 1	$>$ PA
PC1	3.030	27.5	27.5	1.064	Yes	Yes
PC2	2.494	22.7	50.2	1.048	Yes	Yes
PC3	1.556	14.1	64.4	1.034	Yes	Yes
PC4	0.971	8.8	73.2	1.021	No	No
PC5	0.720	6.5	79.7	1.011	No	No
PC6	0.607	5.5	85.3	0.999	No	No
PC7	0.523	4.8	90.0	0.989	No	No
PC8	0.502	4.6	94.6	0.978	No	No
PC9	0.337	3.1	97.6	0.966	No	No
PC10	0.228	2.1	99.7	0.953	No	No
PC11	0.033	0.3	100.0	0.937	No	No

Note. PA = parallel analysis (95th percentile of eigenvalues from random data, 100 iterations, seed = 123). Analysis based on the correlation matrix of all 6,497 wines.

As a sensitivity check, the PCA was repeated without the 175 multivariate outliers flagged in Section 5.1. The first three components explained 66.6% of the variance (compared with 64.4%), and their loadings were almost identical to the original ones (Tucker

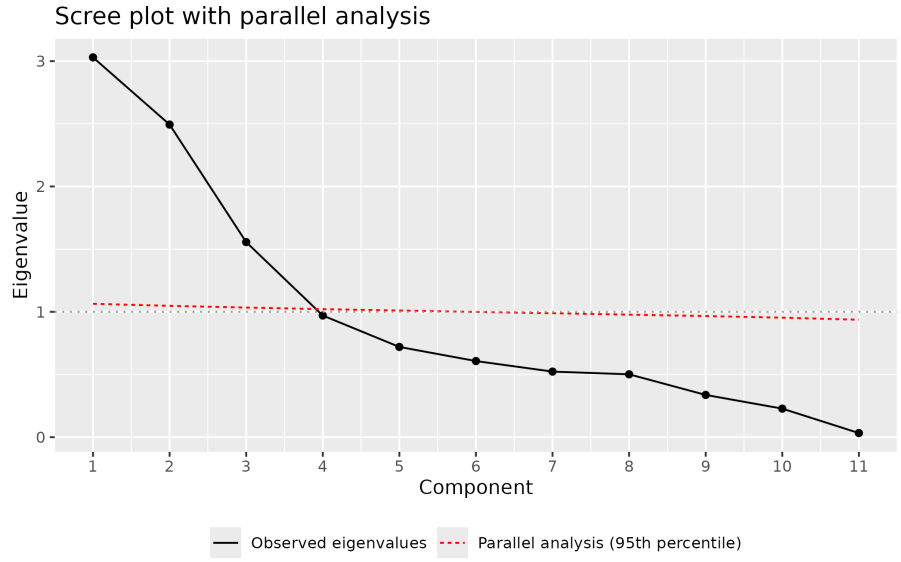


Figure 7: Scree plot of the observed eigenvalues with the parallel-analysis threshold (red dotted line). Three components lie above the threshold.

congruence = 1.00, .99, and .99). The retained solution is therefore not driven by the extreme observations.

5.3.3 Component loadings.

Table 8 reports the loadings (correlations between each variable and each component) and the communalities. Loadings of |.40| or higher are in bold. Signs are arbitrary, so each component was oriented so that its largest loading is positive, and components are described as contrasts. Communalities were high for density (.90), total sulfur dioxide (.76), and fixed acidity (.75), but lower for sulphates (.36) and chlorides (.50), which means that three components reproduce these two variables only partly.

5.3.4 PC1: Sulfur dioxide and sugar versus volatile acidity and salts.

PC1 (27.5% of the variance) has its largest positive loadings on total sulfur dioxide (.85), free sulfur dioxide (.75), and residual sugar (.60), and its negative loadings on volatile acidity (−.66), sulphates (−.51), chlorides (−.50), and fixed acidity (−.42). Total and free sulfur dioxide (24% and 19%) and volatile acidity (15%) contribute most to its definition. Sulfur dioxide is the preservative that protects wine against oxidation and microbial spoilage, and it is commonly used more heavily in wines that retain sugar; PC1 thus opposes a preserved, sugar-retaining profile to a profile of higher volatile acidity and mineral-salt content. This component is essentially the red–white contrast. In the PC1–PC2 biplot (Figure 8) the red wines lie on the left and the white wines on the right with little overlap, and the mean PC1 score was −2.51 for red and 0.82 for white wines (Cohen's $d = -3.39$). The component was computed without using wine color, so the fact that it recovers the grouping agrees with the large MANOVA effect in Section 5.3

Table 8: Component Loadings and Communalities for the Three Retained Components

Variable	PC1	PC2	PC3	h^2
Fixed acidity	**−0.42**	**0.53**	**0.54**	.75
Volatile acidity	**−0.66**	0.19	−0.38	.62
Citric acid	0.27	0.29	**0.74**	.70
Residual sugar	**0.60**	**0.52**	−0.21	.68
Chlorides	**−0.50**	**0.50**	−0.02	.50
Free sulfur dioxide	**0.75**	0.11	−0.17	.60
Total sulfur dioxide	**0.85**	0.14	−0.13	.76
Density	−0.08	**0.92**	−0.22	.90
pH	−0.38	−0.25	**−0.57**	.53
Sulphates	**−0.51**	0.30	0.09	.36
Alcohol	−0.19	**−0.73**	0.33	.68
Eigenvalue	3.03	2.49	1.56	
% Variance	27.5	22.7	14.1	

Note. h^2 = communality (sum of squared loadings over the three components). Bold = |loading| $\geq$ .40.

and reflects the pooled composition of the data.

5.3.5 PC2: Density and extract versus alcohol.

PC2 (22.7%) is defined mainly by density (.92) and alcohol (−.73), with smaller positive loadings for fixed acidity (.53), residual sugar (.52), and chlorides (.50). Density (34%) and alcohol (22%) account for over half of its content. The component contrasts wines whose density is raised by dissolved sugar, acids, and salts with wines whose higher alcohol lowers their density, because ethanol is less dense than water; PC2 can thus be read as a body (density–extract) dimension. Red and white wines overlap substantially on this component (mean scores 0.78 vs. −0.25, $d = 0.68$, a medium difference), so it describes variation that is only partly tied to wine type.

5.3.6 PC3: Fixed-acid content versus pH.

PC3 (14.1%) is defined by citric acid (.74), fixed acidity (.54), and pH (−.57), with a sub-threshold loading for volatile acidity (−.38). Citric acid (35%), pH (21%), and fixed acidity (19%) contribute most. High scores correspond to wines rich in citric and fixed acids and low in pH, which is the expected chemical relation between acid content and pH; this is an acidity-strength (freshness) dimension distinct from the volatile-acidity pole of PC1. The color difference on PC3 was negligible ($d = -0.20$), although red wines were more dispersed than white wines ($SD = 1.77$ vs. 1.02), so this dimension applies to both types.

Three components are sufficient to summarize the physicochemical characteristics of the wines with moderate loss of information: they replace 11 correlated variables with

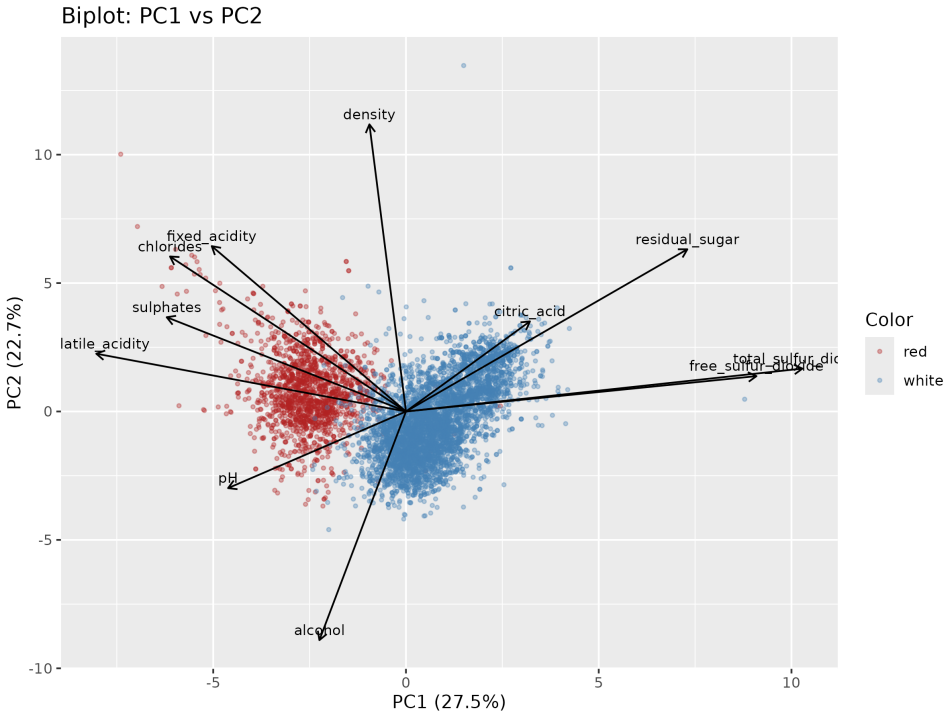


Figure 8: Biplot of PC1 and PC2. Points are wines (red = red wines, blue = white wines); arrows show the variable loadings, scaled to the score range.

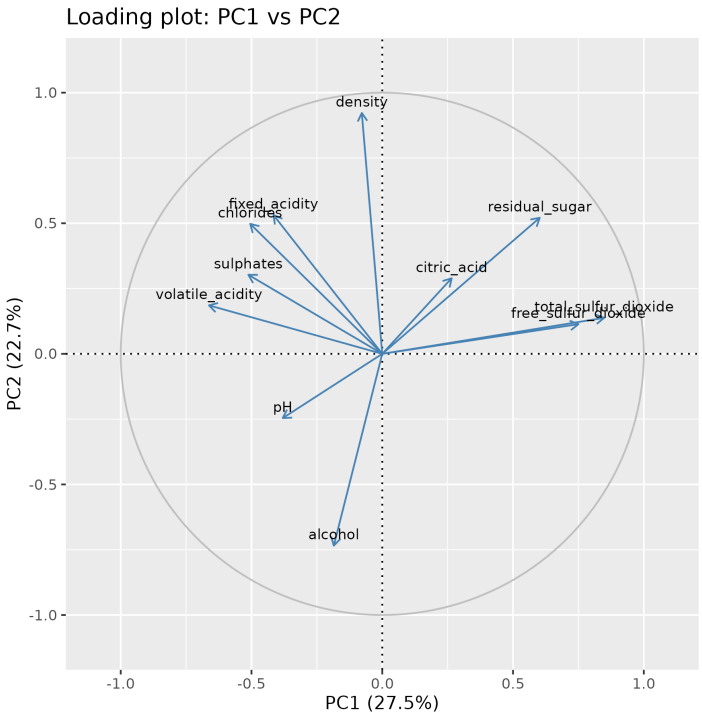


Figure 9: Loading plot of the variables on PC1 and PC2. Variables near the unit circle are well represented by the plane of the two components.

three dimensions that retain 64.4% of the total variance (66.6% when multivariate outliers are removed), leaving roughly a third of the variance unexplained. The dimensions have clear chemical meaning: a preservative–sugar versus volatile-acidity–salts contrast that coincides with wine type (PC1), a density–extract versus alcohol dimension (PC2), and a fixed-acid versus pH dimension (PC3). Two limitations apply. First, because red and white wines are pooled, PC1 largely reflects wine type, and the structure of PC2 and PC3 within each type was not examined separately. Second, the low KMO (.405) means that the components are best treated as data summaries.

5.4 Exploratory Factor Analysis

Factorability

The correlation structure is the one described in Section 5.3.1: 16 of the 55 variable pairs had $|r| \geq .30$ and 3 had $|r| \geq .50$. Bartlett's test rejected the hypothesis that the correlation matrix is an identity matrix, $\chi^2(55) = 37{,}094.62$, $p < .001$, and the determinant of the correlation matrix was .0033, well above the threshold of .00001 for multicollinearity. The overall KMO, however, was .405, below the .50 floor, and five variables had individual values below .50 (pH = .21, alcohol = .27, fixed acidity = .28, residual sugar = .29, density = .30).

The diagnostic recomputations (Table 9) show where the problem lies. Analyzing red and white wines separately did not help (red = .43, white = .37), so pooling the two wine types does not explain the low KMO. Removing density raised the overall KMO to .66, which is acceptable. Density is closely determined by alcohol, residual sugar, and the acid content of the wine, so its partial correlations with the other variables are large relative to its zero-order correlations, and it drags down the adequacy of the other variables it overlaps with. The data are therefore only marginally suitable for factor analysis in their current form: the variables are clearly intercorrelated, but one variable is nearly a linear combination of others. As planned, the factor model was nevertheless fitted to all 11 variables, and the consequences of retaining density are examined below.

Table 9: Overall Kaiser–Meyer–Olkin Measure Under Different Subsets of the Data

Analysis	Overall MSA
All wines, 11 variables	.405
Red wines only	.432
White wines only	.371
All wines, density removed (10 variables)	.663

Note. MSA = measure of sampling adequacy. Values above .60 were considered acceptable.

Number of factors

The criteria did not all agree (Table 10, Figure 10). Parallel analysis for common factors suggested four factors: the reduced-matrix eigenvalues of the first four factors (2.46,

1.58, 0.68, 0.22) exceeded the 95th percentile of the random data (0.36, 0.05, 0.04, 0.03), whereas the fifth was negative and fell below its threshold. Velicer's MAP test was smallest for two factors (.0723, compared with .0734 for one and .0763 for three). The eigenvalue-greater-than-one rule applied to the full correlation matrix, which was reported for reference only, indicated three (3.03, 2.49, 1.56; the fourth was 0.97). Among the solutions compared, the three-factor solution had the lowest RMSEA (.252) and the highest fit index (.83), while BIC decreased steadily as factors were added. Because BIC with $N = 6{,}497$ rewards additional factors almost regardless of their value, it was not used as the deciding criterion.

Table 10: Eigenvalues, Parallel-Analysis Thresholds, and Fit Indices for Different Numbers of Factors

Factors	Eigenvalue (full **R**)	Eigenvalue (reduced)	PA threshold	MAP	RMSEA	BIC
1	3.030	2.460	0.356	.0734	.304	26,100
2	2.494	1.583	0.051	.0723	.257	14,293
3	1.556	0.680	0.039	.0763	.252	10,086
4	0.971	0.221	0.028	.0976	.280	8,545
5	0.720	−0.043	0.017	.1148	.319	6,524
6	0.607	−0.100	0.006	.1686	.402	4,168

Note PA = parallel analysis (95th percentile of eigenvalues from 100 random data sets, seed = 123; minimum residual). MAP = Velicer's minimum average partial statistic (smallest at two factors). RMSEA and BIC are from minres solutions with oblimin rotation.

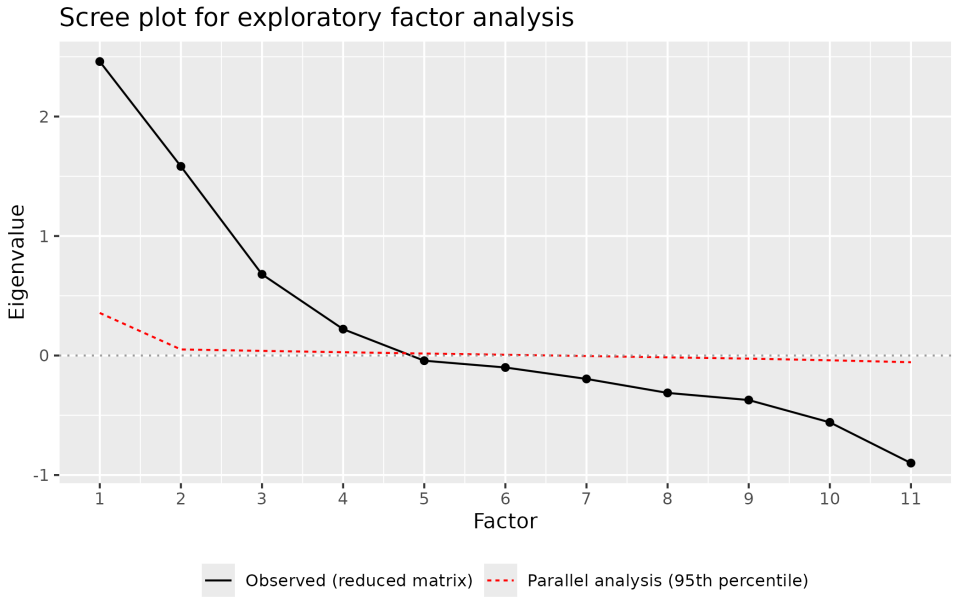


Figure 10: Scree plot of the reduced-matrix eigenvalues for the exploratory factor analysis, with the parallel-analysis threshold (red line).

The solutions with two, three, and four factors were then compared directly (Table 11). Going from two to three factors halved the RMSR (.102 to .055), raised the mean communality from .42 to .53, and reduced the number of variables without a salient loading

from three to two. A fourth factor lowered the RMSR further (.040) but made RMSEA worse (.280) and TLI worse (.389 to .242), and it did not reduce the number of cross-loading variables (four in both solutions). The three-factor solution was therefore retained as the best compromise between the two-factor suggestion of the MAP test and the four-factor suggestion of parallel analysis. It is also the solution that reproduces the three-dimensional structure found in the PCA and has a clear chemical reading. Parallel analysis is known to over-extract in large samples, so its fourth factor was treated with caution. The fit indices of the retained model are poor by conventional standards (RMSR = .055, RMSEA = .252, 90% CI [.248, .256], TLI = .389) and are reported as descriptive indices only, since RMSEA and TLI are sensitive to the sample size and to the Heywood case described below.

Table 11: Comparison of the Two-, Three-, and Four-Factor Solutions

Factors	RMSR	RMSEA	TLI	BIC	Mean h^2	No salient loading	Cross-loading
2	.102	.257	.364	14,293	.423	3	3
3	.055	.252	.389	10,086	.531	2	4
4	.040	.280	.242	8,545	.582	1	4

Note. Minimum residual extraction, oblimin rotation. "No salient loading" counts variables with no loading of |.40| or higher; "Cross-loading" counts variables with loadings of |.30| or higher on more than one factor. All three solutions produced an ultra-Heywood case (density).

Extraction and rotation

Factors were extracted by minimum residual (minres) and rotated with oblimin. The estimated factor correlations were negligible: the largest was between the body and acid-strength factors ($\phi = .17$), and the other two were $-.02$ and .01. The oblique rotation therefore reduced in practice to an orthogonal solution, which is the outcome anticipated in Section 4.5. Promax rotation and maximum likelihood extraction gave essentially the same factors (Tucker's congruence .88–.98; Table 13).

Factor loadings, communalities, and uniquenesses

Table 12 reports the pattern matrix, communalities, and uniquenesses of the three-factor solution, which together explain 53.1% of the total variance (23.2%, 18.9%, and 10.9% for the three factors). Figure 11 displays the pattern matrix.

Most variables loaded clearly on one factor: total and free sulfur dioxide on F1, density and alcohol on F2, and citric acid and pH on F3. Four variables cross-loaded ($|\text{loading}| \geq .30$ on more than one factor): volatile acidity (on all three factors), fixed acidity (F1 and F3), residual sugar (F1 and F2), and chlorides (F1 and F2). Chlorides and sulphates had no loading of |.40| or higher and are the weakest variables, with communalities of .32 and .22, and pH was also poorly reproduced ($h^2 = .29$). The three factors therefore account well for sulfur dioxide, density, fixed acidity, and citric acid, but not for the salt measures (chlorides and sulphates) or pH.

Table 12: Pattern Matrix, Communalities, and Uniquenesses for the Three-Factor Solution

Variable	F1	F2	F3	h^2	u^2
Fixed acidity	−.**43**	.27	**.61**	.70	.30
Volatile acidity	−.**55**	.34	−.32	.49	.51
Citric acid	.19	−.06	**.69**	.51	.49
Residual sugar	**.60**	**.45**	.02	.55	.45
Chlorides	−.39	.39	.05	.32	.68
Free sulfur dioxide	**.70**	.07	−.02	.49	.51
Total sulfur dioxide	**.86**	.08	.01	.74	.26
Density	.00	**1.01**	.04	1.04	−.04
pH	−.28	.03	−.**47**	.29	.71
Sulphates	−.40	.22	.08	.22	.78
Alcohol	−.23	−.**66**	.05	.48	.52
SS loadings	2.56	2.08	1.20		
% variance	23.2	18.9	10.9		

Note Minres extraction, oblimin rotation, N = 6,497. Bold = |loading| ≥ .40. The loadings of −.39 (chlorides) and −.40 (sulphates) on F1 are just below the .40 cutoff before rounding. Factor correlations: F1–F2 = −.02, F1–F3 = .01, F2–F3 = .17. F1 = Preservation versus volatile acidity and salts; F2 = Body (density–extract) versus alcohol; F3 = Fixed-acid strength.

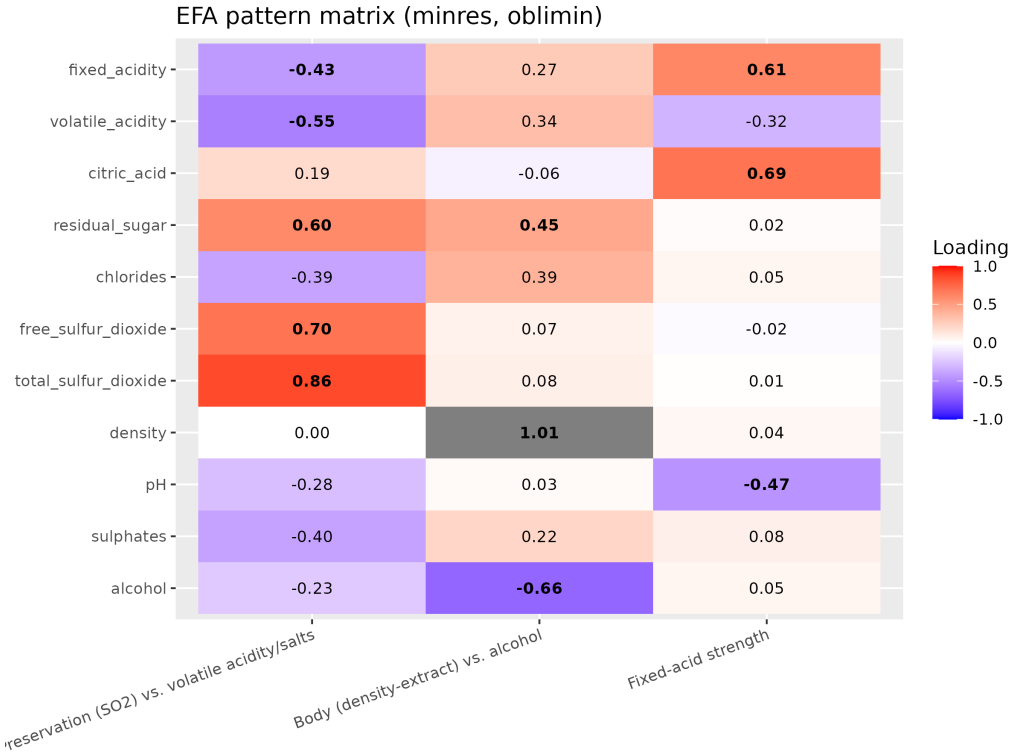


Figure 11: Pattern matrix of the three-factor solution (minres, oblimin). Bold values have |loading| ≥ .40.

Density had a communality of 1.04 (uniqueness −.04), an ultra-Heywood case, and the software warned that the factor-score weights were probably unreliable. The flag appeared in every solution examined and means that F2 is almost entirely a density factor: the common factors reproduce all of the variance of density, so the model cannot be trusted to apportion its variance exactly. This is consistent with the very high VIF of density in the MANOVA variable selection (15.9) and with its low KMO. As a sensitivity check, the model was refit without density. The Heywood case disappeared, the overall KMO rose to .66, and three factors explained 47.5% of the variance (RMSR = .048, RMSEA = .139, TLI = .704). The fixed-acid factor was reproduced exactly (Tucker's congruence = 1.00) and the preservation factor closely (.94). The density factor was reproduced less well (.88): without density, it became a contrast between volatile acidity, chlorides, and sulphates on one side and alcohol on the other, which is the part of F2 that did not belong to density itself. The interpretation below is therefore based on the 11-variable solution, with this caveat.

F1: Preservation versus volatile acidity and salts

F1 (23.2%) has its largest positive loadings on total sulfur dioxide (.86), free sulfur dioxide (.70), and residual sugar (.60), and negative loadings on volatile acidity (−.55), fixed acidity (−.43), and, just below the salient cutoff, sulphates (−.40) and chlorides (−.39). Sulfur dioxide is the preservative that protects wine against oxidation and microbial spoilage, and it is used more heavily in wines that retain sugar. F1 therefore opposes a heavily preserved, sugar-retaining profile to a profile with more volatile acidity (acetic acid, a sign of spoilage) and more mineral salts. It is the red–white factor: mean factor scores were −1.32 for red and 0.43 for white wines, a very large difference ($d = -3.17$).

F2: Body (density–extract) versus alcohol

F2 (18.9%) is defined by density (1.01) and alcohol (−.66), with a smaller contribution from residual sugar (.45) and sub-threshold loadings for chlorides (.39) and volatile acidity (.34). Because ethanol is less dense than water, wines with high alcohol have low density, whereas dissolved sugar and other extract raise it. F2 therefore describes the body of the wine: dense, extract-rich, low-alcohol wines at one end and light, high-alcohol wines at the other. Red wines scored higher on this factor than white wines (0.87 vs. −0.28, $d = 1.06$), but with considerable overlap.

F3: Fixed-acid strength

F3 (10.9%) is defined by citric acid (.69), fixed acidity (.61), and pH (−.47), with a sub-threshold loading for volatile acidity (−.32). Wines high on F3 are rich in citric and fixed acids and have a lower pH, which is the expected chemical relation between acid content and acidity strength. Red and white wines did not differ on this factor ($d = -.04$), although red wines were much more dispersed ($SD = 1.36$ vs. 0.63), so this dimension applies to both wine types.

Robustness

The solution was compared with alternative solutions using Tucker's congruence coefficient (Table 13). It was essentially unchanged when the 175 multivariate outliers were removed (.98–1.00), and it was close to identical with maximum likelihood extraction and promax rotation for the fixed-acid and body factors (.88–.98), although the preservation factor was somewhat less stable under promax rotation (.88). The pooled structure did not hold as well within wine types. The fixed-acid and body factors were recovered reasonably well in red (.94 and .91) and white wines (.89 and .91), but the preservation factor was not (.76 in red, .66 in white). Fitting the model separately showed why: within each wine type, free and total sulfur dioxide formed a factor of their own (loadings .74–.86), without residual sugar, volatile acidity, or the salts, and in white wines residual sugar loaded instead with density and alcohol (.74, 1.04, and −.69). Part of the pooled F1 therefore reflects the difference between red and white wines, which have different chemical profiles, and not a single dimension that operates within each type. The white-wine solution also produced a Heywood case.

Table 13: Tucker's Congruence Between the Pooled Factors and Alternative Solutions

Comparison	F1 Preservation	F2 Body	F3 Fixed-acid
Promax rotation	.88	.93	.98
Maximum likelihood extraction	.94	.97	.88
Without multivariate outliers	.99	1.00	.98
Red wines only	.76	.91	.94
White wines only	.66	.91	.89

Note. Each pooled factor is matched to the factor of the alternative solution with which it has the highest congruence. Values of .95 or higher indicate essentially identical factors.

The relationships among the 11 physicochemical variables can be described by three nearly uncorrelated latent dimensions that explain about 53% of the total variance: a preservation dimension (sulfur dioxide and residual sugar opposed to volatile acidity and salts) that separates red from white wines, a body dimension (density and extract opposed to alcohol), and an acid-strength dimension (citric acid, fixed acidity, and pH). Fit is poor by conventional standards, density is an improper (Heywood) variable, chlorides, sulphates, and pH are only weakly represented, and the first factor is partly a product of pooling the two wine types. The factors are best read as a plausible summary of the chemical structure and not as a confirmed latent model.

6 Integrated Discussion

What distinguishes red and white wines

The MANOVA showed that red and white Vinho Verde wines are chemically distinct (Pillai's trace = 0.79, partial η^2 = .79). The difference is carried mainly by total sulfur

dioxide ($d = -2.28$) and volatile acidity ($d = 2.00$): white wines are heavily preserved and retain more residual sugar, whereas red wines have higher volatile acidity, fixed acidity, chlorides, sulphates, and pH. Alcohol and citric acid separate the two types little or not at all. The unsupervised analyses recovered the same distinction without using wine color. PC1 separated red from white wines with $d = -3.39$ and the preservation factor of the EFA with $d = -3.17$, and both are defined by the same variables that dominate the discriminant function.

Major dimensions of variation

PCA reduced the 11 variables to three components retaining 64.4% of the total variance: a preservative–sugar versus volatile-acidity–salts contrast (PC1, 27.5%), a density–extract versus alcohol dimension (PC2, 22.7%), and a fixed-acid versus pH dimension (PC3, 14.1%). Only the first is tied strongly to wine type. The second differs between red and white wines to a moderate degree ($d = 0.68$) and the third not at all ($d = -0.20$), so most of the variation along PC2 and PC3 is shared by both types.

Latent factors

The EFA identified three nearly uncorrelated factors explaining 53.1% of the variance: preservation versus volatile acidity and salts, body (density and extract) versus alcohol, and fixed-acid strength. Sulfur dioxide, density, alcohol, citric acid, and fixed acidity define the factors well, whereas chlorides, sulphates, and pH are only weakly represented.

PCA and EFA compared

The two methods gave nearly the same picture. The components and factors correspond one to one in order, composition, and size, and the loadings of the main variables are similar (total sulfur dioxide .85 vs. .86, residual sugar .60 vs. .60, alcohol −.73 vs. −.66, citric acid .74 vs. .69, pH −.57 vs. −.47). The agreement is helped by the fact that the factor correlations were close to zero ($|\phi| \leq .17$), so the oblique EFA solution is close to the orthogonal PCA one.

The differences are mostly in what each method models. PCA summarizes the total variance of the variables, so the three components retain more variance (64.4%) than the factors explain of the common variance (53.1%, or 47.5% without density). Its loadings are generally larger, particularly for the variables with substantial unique variance such as chlorides (−.50 vs. −.39) and sulphates (−.51 vs. −.40). EFA removes the unique variance, and the resulting loadings expose the weakness of those variables more clearly: their communalities fell to .32 and .22, against .50 and .36 in the PCA. EFA also exposed the redundancy of density, which produced a Heywood case ($h^2 = 1.04$) that PCA absorbs without difficulty (.90). PCA makes no claim about latent constructs, and its low KMO (.405) was not a barrier to its use as a data summary. For EFA, which does posit latent factors, the same KMO is a real weakness.

Variables that are important in both analyses are total and free sulfur dioxide, density, alcohol, citric acid, and fixed acidity, each of which loaded strongly on the same dimension in both. Volatile acidity and residual sugar are also prominent in both, although they cross-load in the EFA. Chlorides and sulphates are secondary in both.

Substantive interpretation

Taken together, the three analyses show that the chemistry of Vinho Verde wines is organized around three themes: how heavily the wine is preserved and how much sugar it retains, how dense (extract-rich) or alcoholic it is, and how strong its fixed acidity is. Red and white wines differ mainly on the first theme, which is the only one that reflects a production difference between the two types, partly overlap on the second, and are indistinguishable on the third, except that red wines vary more along it. White wines are marked by high sulfur dioxide and sugar, and red wines by high volatile acidity and mineral salts. In practice, this means that total sulfur dioxide and volatile acidity are the most informative measures for telling the wine types apart, whereas alcohol and citric acid tell them little about the type.

Limitations

1. **Observational data and independence.** The wines come from a single observational data set with no producer, batch, or vintage identifiers, so independence could not be verified. Duplicate rows (18.1%) could not be distinguished from genuinely repeated measurements and were retained.

2. **Unmet MANOVA assumptions.** Multivariate normality was rejected in both wine types and the covariance matrices differed considerably (Box's M). Pillai's trace, the Welch tests, and the very large effect make the central conclusion unlikely to change, but the exact F and p values should be interpreted cautiously.

3. **Pooling of two wine types.** PC1 and the first factor largely reproduce the red–white contrast, and the within-type EFAs showed that the preservation factor is partly an artifact of that mixture. The group sizes are also unbalanced (24.6% red, 75.4% white), which gives the pooled correlation structure a stronger white-wine character.

4. **Weak factorability and an improper solution.** The overall KMO was .405 and density produced a Heywood case, so the factor-score weights and the exact loadings are unreliable. The fit of the factor model was poor (RMSEA = .252, TLI = .389), and the factors were only partly stable across extraction method, rotation, and wine type.

5. **Skewed variables and outliers.** Pearson correlations, on which the PCA and EFA rest, are sensitive to the heavy tails of chlorides, residual sugar, sulphates,

and free sulfur dioxide. The sensitivity analyses suggested that the retained 175 multivariate outliers did not alter the solutions materially.

6. **Scope.** The wines are all Vinho Verde from northern Portugal, and the sensory quality score was not analyzed, so the findings describe the chemistry of these wines and not what makes a wine good. The same sample was used to explore and interpret the structure, and it was not cross-validated, so the latent factors should be regarded as exploratory.

7 Conclusion

Red and white Vinho Verde wines differ markedly in their physicochemical profile. The MANOVA was significant with a very large effect (Pillai's trace = 0.79, $F(7, 6489) = 3518.70$, $p < .001$, partial $\eta^2 = .79$), and the difference is driven mainly by total sulfur dioxide, which is higher in white wines, and by volatile acidity, fixed acidity, chlorides, and sulphates, which are higher in red wines. The 11 physicochemical variables can be summarized by three principal components retaining 64.4% of the total variance (preservative–sugar versus volatile-acidity–salts, density–extract versus alcohol, and fixed-acid versus pH), the first of which closely recovers the wine type. Exploratory factor analysis supports a very similar three-factor structure (preservation, body, and fixed-acid strength) that explains 53.1% of the variance, with nearly uncorrelated factors. These conclusions come with caveats: several MANOVA assumptions were not met, the KMO was low and density produced a Heywood case, and the first factor partly reflects the pooling of the two wine types. The structure is therefore best treated as an exploratory description of the chemistry of Vinho Verde wines and not a confirmed latent model.

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